

Homework Assignment 7

- Two two-stage launchers are considered to provide an impulsive velocity rise of 9 km/s to a payload. The details of each launcher is specified below. While you may use MATLAB or other utility to solve the problem, the various steps should be described as if you are manually carrying out the solution.

Launcher 1:

Stage 1: $I_{sp1} = 290$ s, $\epsilon_1 = 0.07$ (LOX-RP1)

Stage 2: $I_{sp2} = 435$ s, $\epsilon_2 = 0.10$ (LOX-LH2)

Launcher 2:

Stage 1: $I_{sp1} = 290$ s, $\epsilon_1 = 0.05$ (LOX-N2H4)

Stage 2: $I_{sp2} = 435$ s, $\epsilon_2 = 0.10$ (LOX-LH2)

Determine, for each launcher, the maximum possible (M_L/M_{01}), and the associated values of the impulsive velocity rise provided by each stage. {Partial Ans.: Launcher 1: $\lambda_1 = 0.5568$, Launcher 2: $\lambda_2 = 0.1781$ }

- A three-stage rocket has the following characteristics:

$$I_{sp1} = I_{sp2} = I_{sp3} = 370 \text{ s}$$

$$\epsilon_1 = \epsilon_2 = \epsilon_3 = 0.15$$

$$\Delta u_{b,imp} = 10807 \text{ m/s}$$

Determine the maximum (M_L/M_{01}). If the payload mass (M_L) is 20000 kg, compute the structural mass of the first stage. {0.0175, $M_{s1} \sim 129200$ kg.}

- A three-stage rocket has the following characteristics:

	Stage 1	Stage 2	Stage 3
Initial mass (kg)	150,000	27000	6000
Propellant mass (kg)	112,500	18000	3000
Payload mass (kg)			1538
Specific impulse (s)	420	420	420

- Calculate the payload ratio and structural coefficient for each stage. Also determine the impulsive velocity rise to the payload. {Ans.: 13094 m/s}
- Assume, now, that the 3-stages are reconfigured to have the same structural coefficient. First, calculate the structural coefficient. Then, optimize the rocket (maintaining the value of $\Delta u_{b,imp}$), and find $(M_L/M_{01})_{max}$. {Ans.: 0.0205}